8. The specialized chamber used for the counting of microbial cells is
   a. Haemocytometer
   b. Counting chamber
   c. Petroff Hauss chamber
   d. Counting slide

Answer the following

1. Give notes on the nutritional classes of microorganisms.
2. Classify microorganisms based on energy and carbon source.
3. What are light and dark reactions in photosynthesis?
4. What is bacteriochlorophyll? Give its role.
5. Define chemoautotroph.
6. Define photosynthesis.
7. Give examples of photosynthetic bacteria.
8. What do mean by cardinal temperature?
9. Give notes on photosynthetic pigments.
10. What are halophiles?
11. Give reason for the ability of thermophiles to grow in high temperatures.
12. How bacterial cells are counted using counting chamber?
13. Classify microorganisms based on their temperature requirement.
14. Describe the role of macro and micronutrients in microorganisms. How do you think bacteria acquire their nutrients from their environment?
15. Explain the classification of microbes based on their nutrition. If H_2S is toxic to living organisms, how do purple and green bacteria survive and use H_2S in such environments.
16. Describe the photosystems of cyanobacteria.
17. Draw a schematic representation of Z scheme of non cyclic photophosporylation.
18. Compare photosynthesis between plants, cyanobacteria and purple green bacteria.
19. Explain the principle and uses of chemostat and turbidostat with diagrams.
20. Describe the classification of microorganism based on their oxygen requirement.
21. Explain the principle and uses of chemostat and turbidostat with diagrams.
22. Explain the relation of osmosis to water activity.
23. Define growth. Explain the phases of growth of bacteria with neat diagram.

Student Activity

- Expose a container with water to sunlight for a week. Observe the growth of cyanobacteria on water which explains the photoautotrophic mode of nutrition.
- Store a loaf of bread for a week after the expiry date. You can observe the growth of fungi/molds which demonstrates the mode of nutrition of chemoheterotrophs.
- Collect rusted iron pipes which contain chemolithotrophic *Thiobacillus sp* which can oxidize iron for their nutrients.
- Place two bowls of cooked rice/vegetables–one inside the refrigerator at 6°C, and another at room temperature at 30-35°C. Give reasons for the quick spoilage of the rice stored at 30-35°C. Check the pH of milk using a pH paper.

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